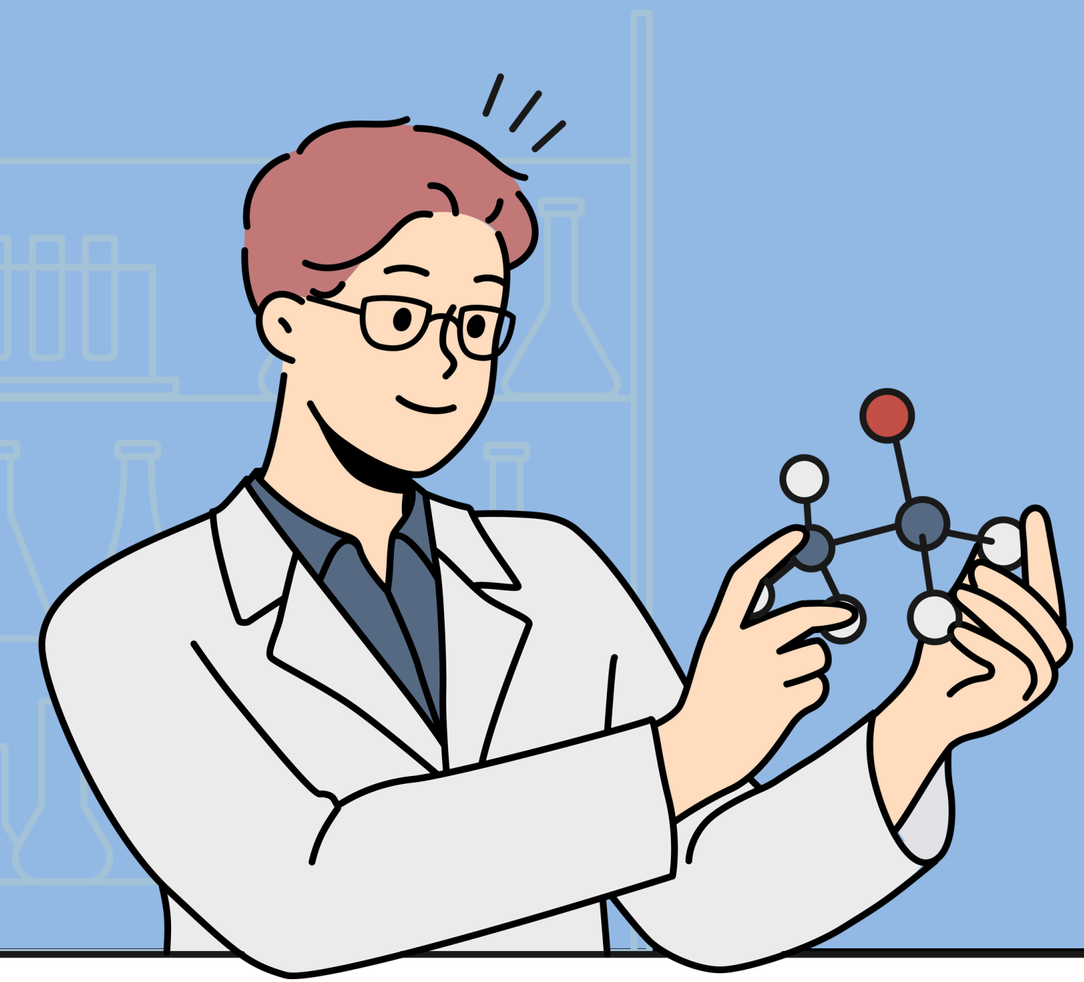
GENERAL CHEMISTRY 2

LEARNING OBJECTIVES



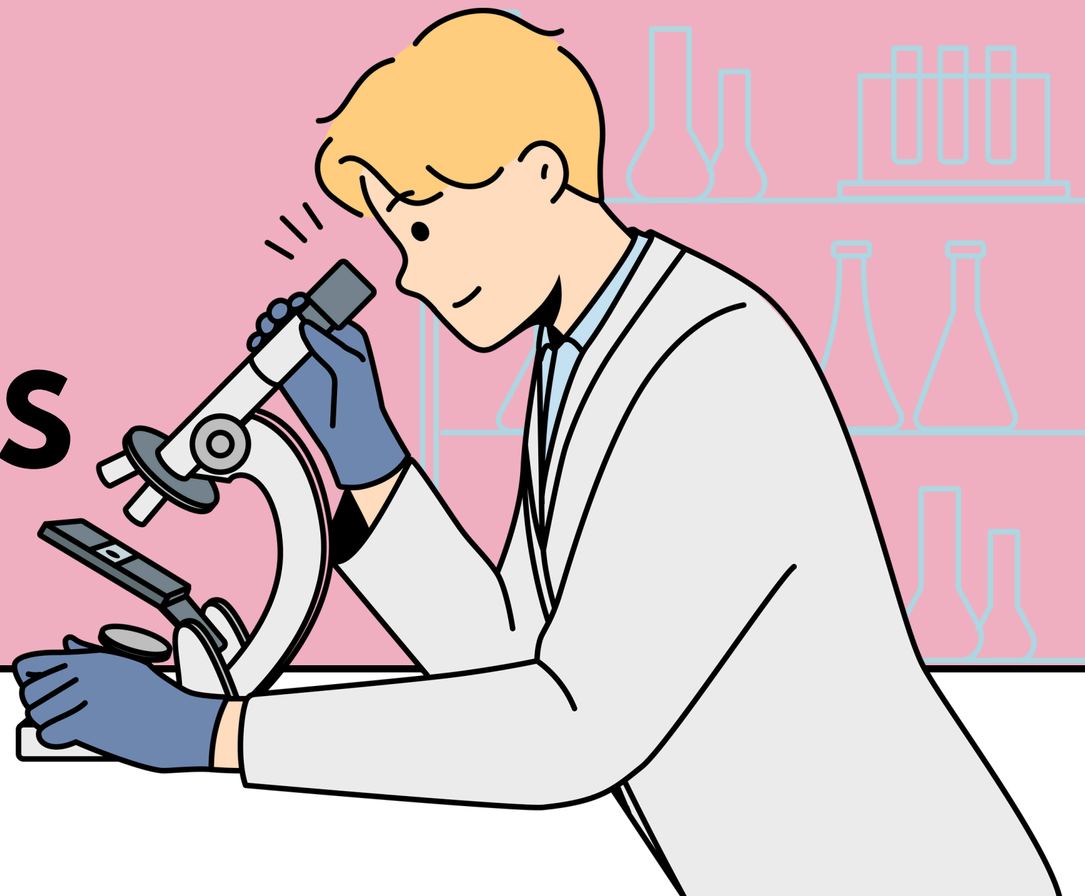
- Use the kinetic molecular model to explain the properties of liquids and solids (STEM_GC11IMFIIIa-c-99)

- Describe and differentiate the types of intermolecular forces (STEM_GC11IMFIIIa-c-100)

- Describe the following properties of liquids, and explain the effect of intermolecular forces on these properties: surface tension, viscosity, vapor pressure, boiling point, and molar heat of vaporization (STEM_GC11IMFIIIa-c-102)

LESSON 1:

THE KINETIC MOLECULAR MODEL AND THE PROPERTIES OF LIQUIDS AND SOLIDS



Directions: For each state of matter, draw the arrangement of particles.

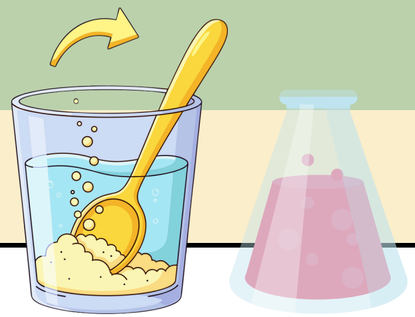
solid

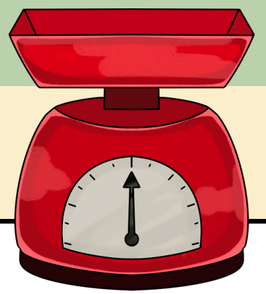
liquid

gas

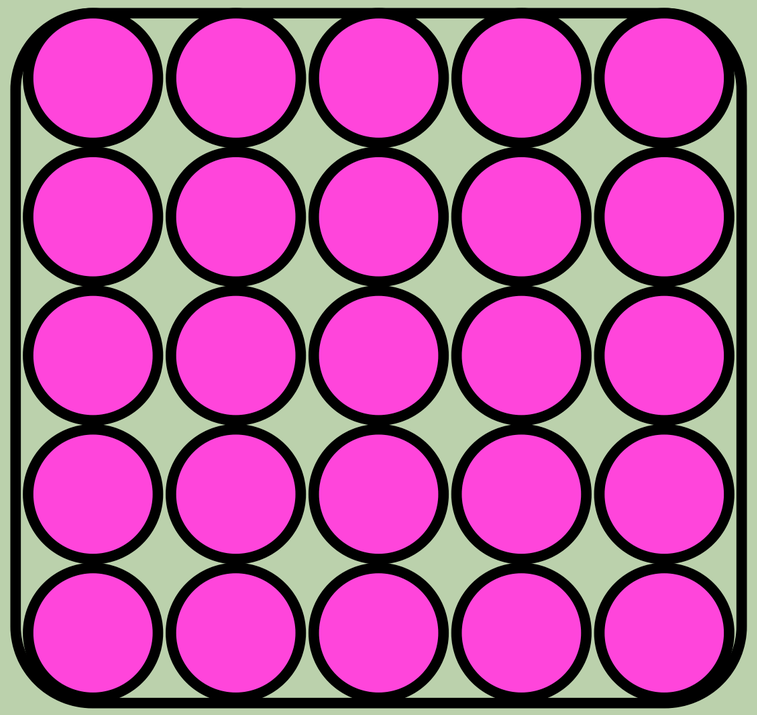
Kinetic Molecular Theory

Is composed of tiny particles that are always in motion. The theory helps explain the observable properties and behaviors of solids, liquids, and gases.

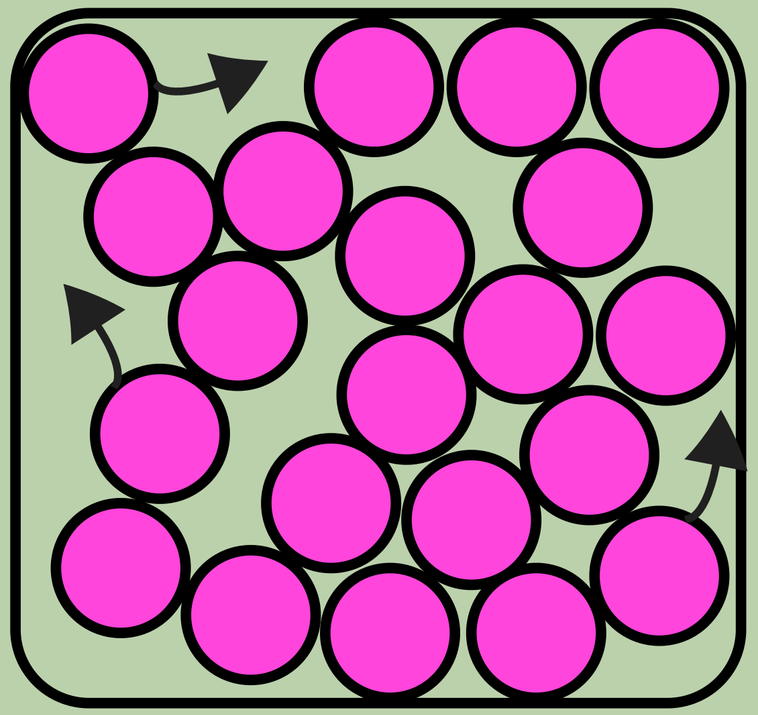




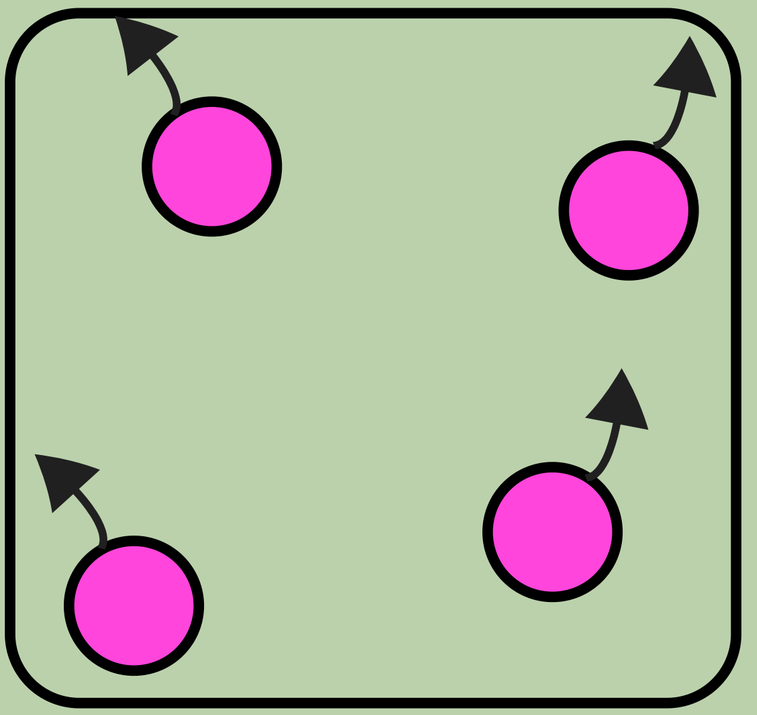
Kinetic Molecular Theory



solid



liquid



gas

ACTIVITY #1

Directions: Complete the table below.

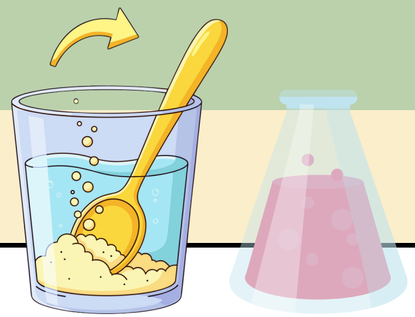
Properties of Matter	Molecular Behavior		
	Gas	Liquid	Solid
Volume/Shape			
Density			
Compressibility			
Motion of Molecules			

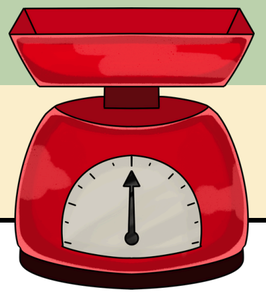
LESSON 2

INTERMOLECULAR FORCES OF ATTRACTION

Intramolecular Forces

a force that exists between two atoms within a molecule.





Intramolecular Forces

Types of Bonds:

Ionic Bond

- Electrons are transferred from one atom to another.
- A metal and a nonmetal.
- Formation of ions (positive and negative) that attract each other.
- Example: NaCl (sodium chloride)

Intramolecular Forces

Types of Bonds:

Covalent Bond

- Atoms share electrons.
- Two nonmetals.
- Formation of molecules.
- Examples: H_2O, CO_2

Intramolecular Forces

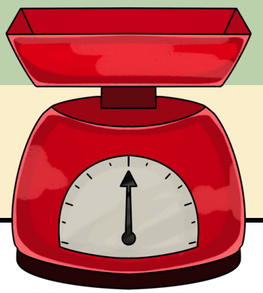
Types of Bonds:

Metallic Bond

- Found in metals.
- Atoms share a “sea of electrons” that move freely.
- Explains why metals conduct heat and electricity.
- Example: Copper, iron, gold

Intermolecular Forces

are attractive forces between molecules or particles in the solid or liquid states.



Intermolecular Forces

The intermolecular forces of attraction in a pure substance are collectively known as Van Der Waals forces.

Intermolecular Forces

Ion-ion

electrostatic attraction between positively charged ions (cations) and negatively charged ions (anions).

Ion-Ion

$$F = \frac{k q_1 q_2}{R^2}$$

$Na^{+} \cdots Cl^{-}$

$Ca^{+2} \cdots O^{-2}$

$$LE \propto \frac{k q_1 q_2}{R}$$

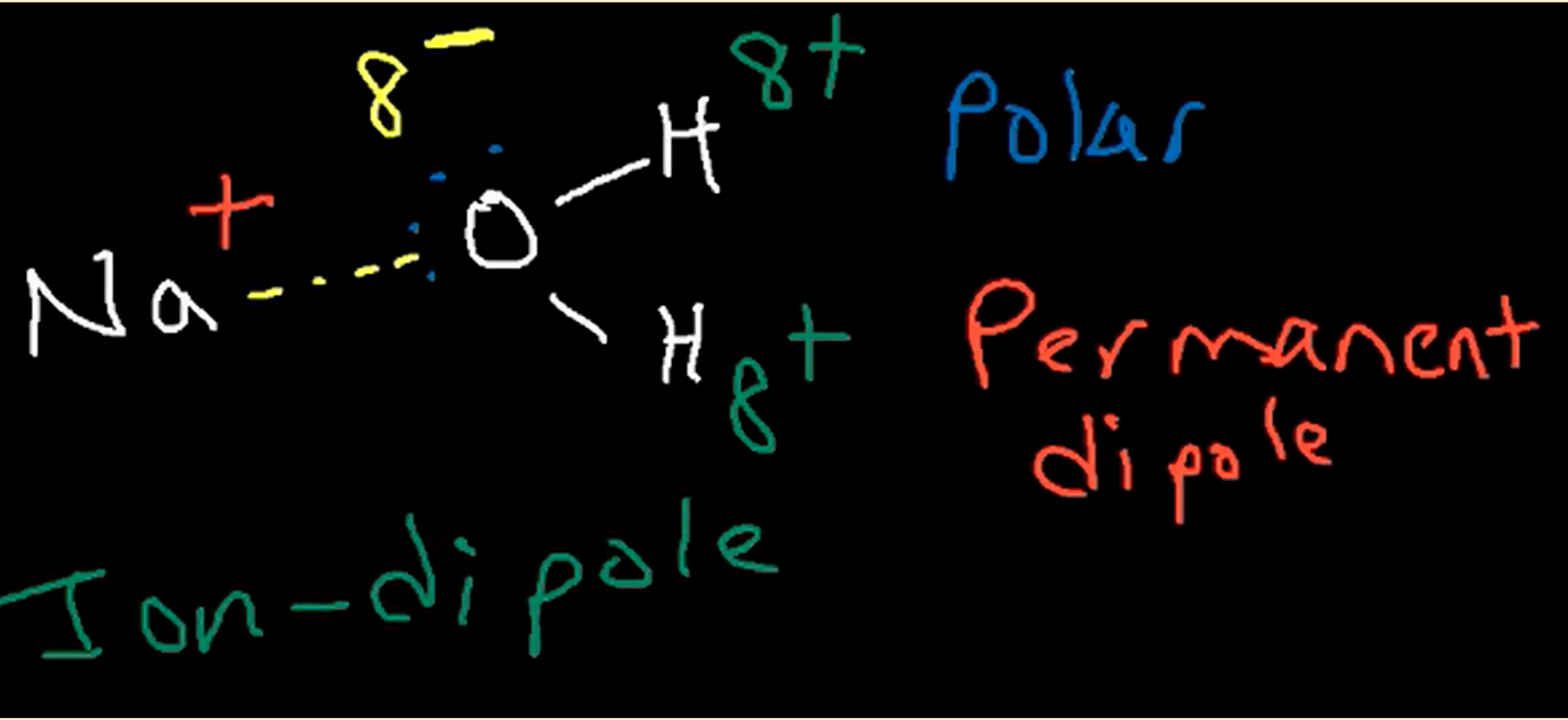
charge ↑ F ↑

size ↑ F ↓

Intermolecular Forces

Ion-dipole

Interaction between ion and a polar molecule.

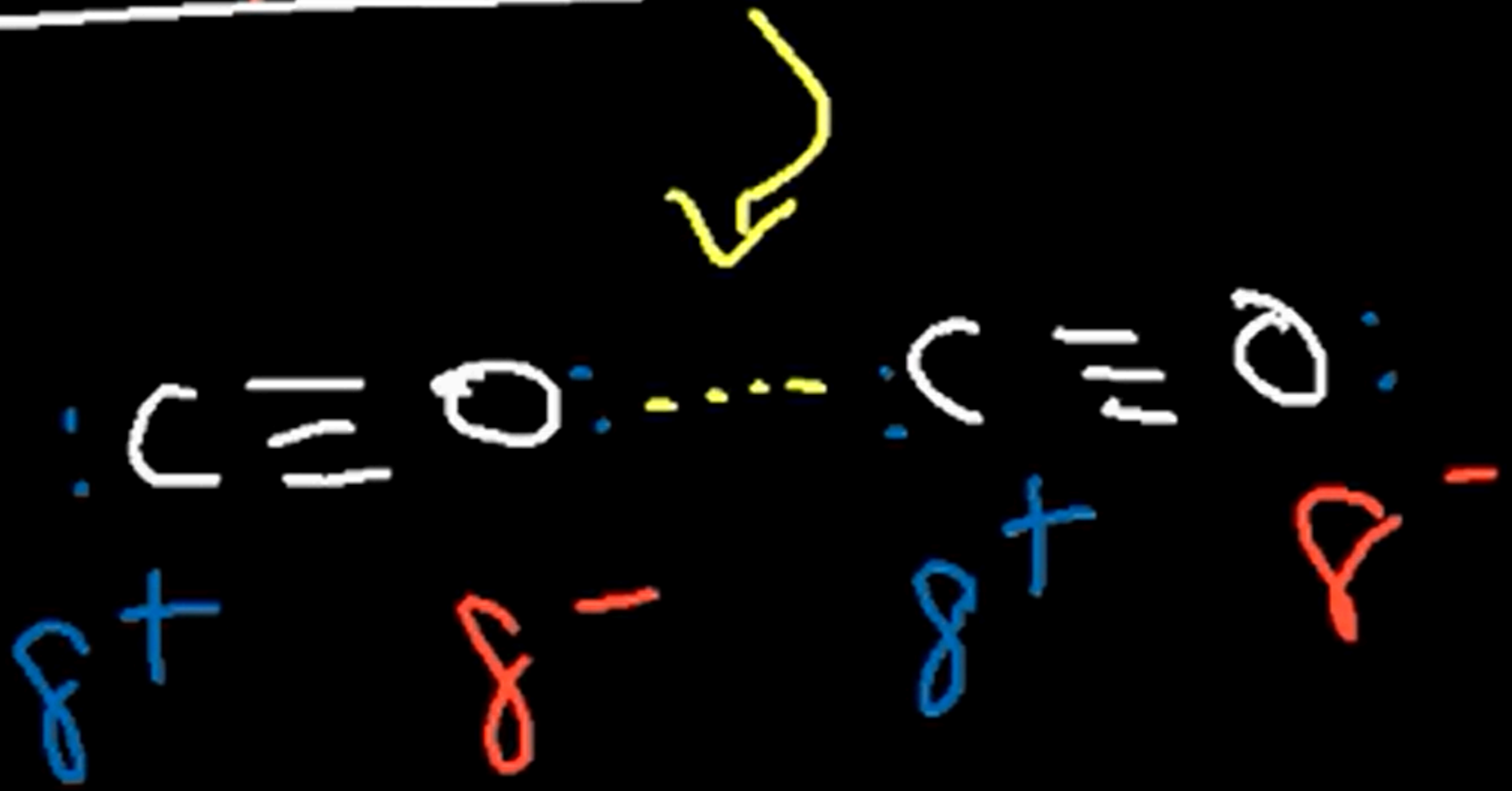

δ−
δ+
Na+
O
H
H
δ+
Polar
Permanent dipole
Ion-dipole

Intermolecular Forces

Dipole-dipole

Exists between polar molecules. One end of a dipole attracts the oppositely charged end of the other dipole.

Dipole – Dipole



Intermolecular Forces

Hydrogen bond

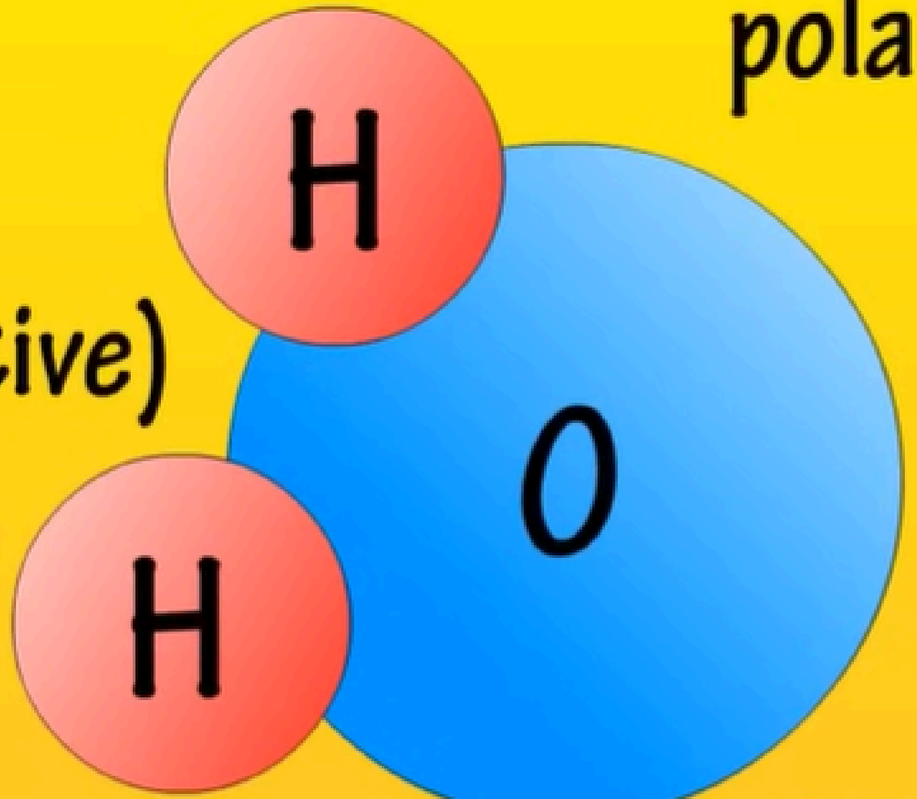
It is a special and very strong type of dipole - dipole force that exists between a hydrogen atom bound to a small and small and highly electronegative non-metal atom.

hydrogen bonds

polar molecule

+

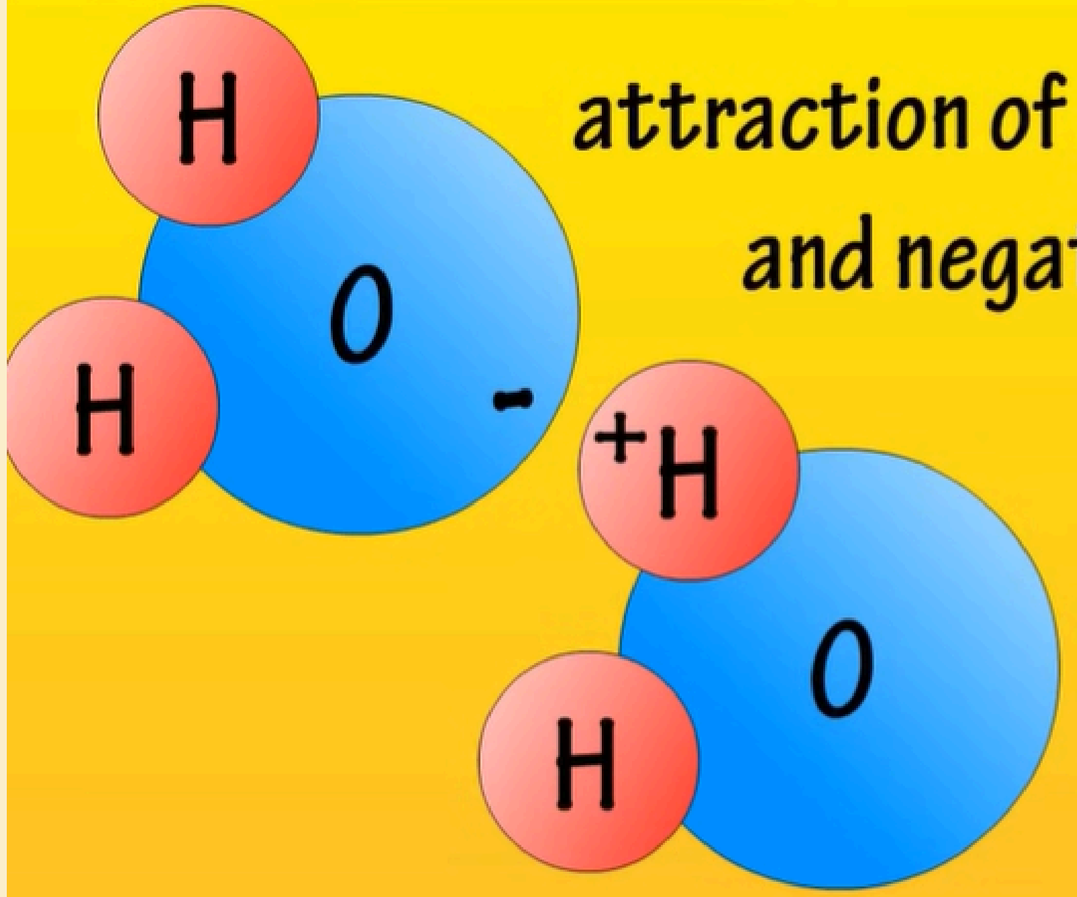
(slightly positive)



-

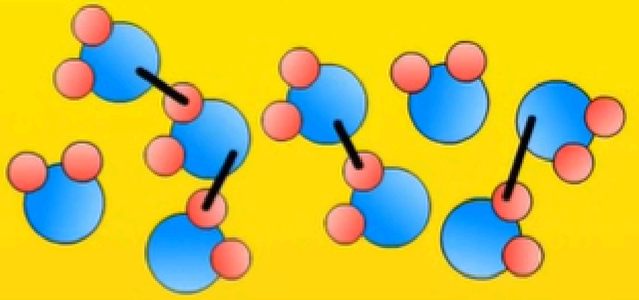
(slightly negative)

hydrogen bonds



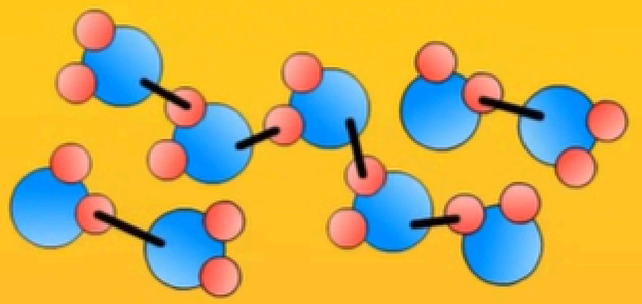
attraction of positive and negative

liquid



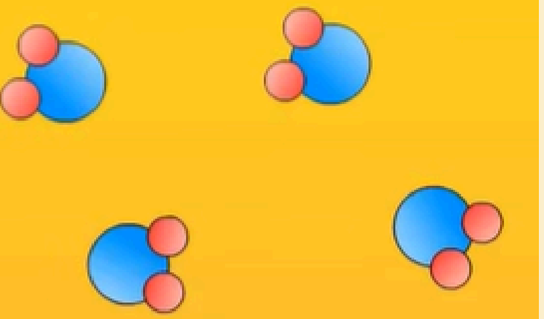
a few bonds

solid form

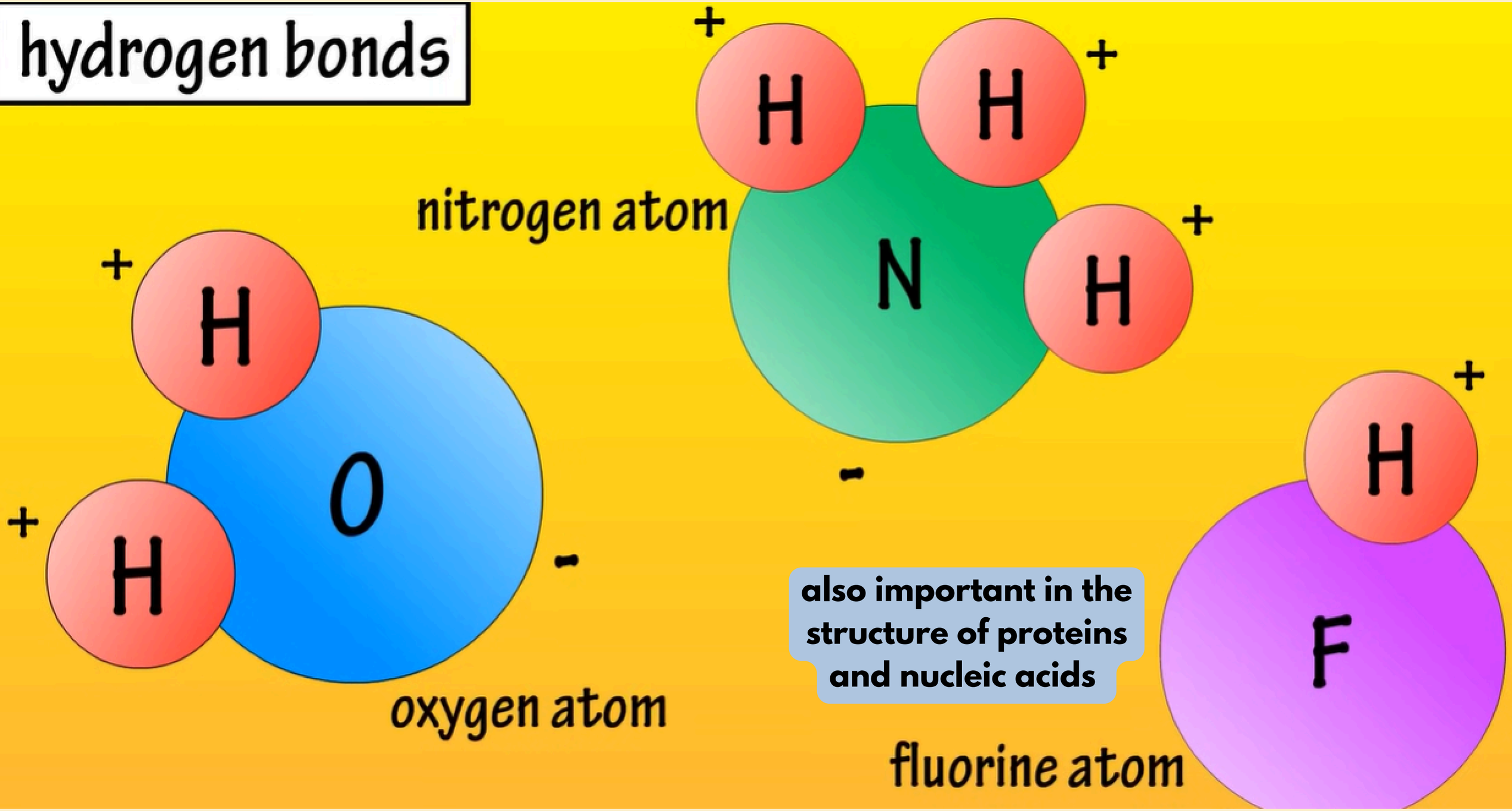


many bonds

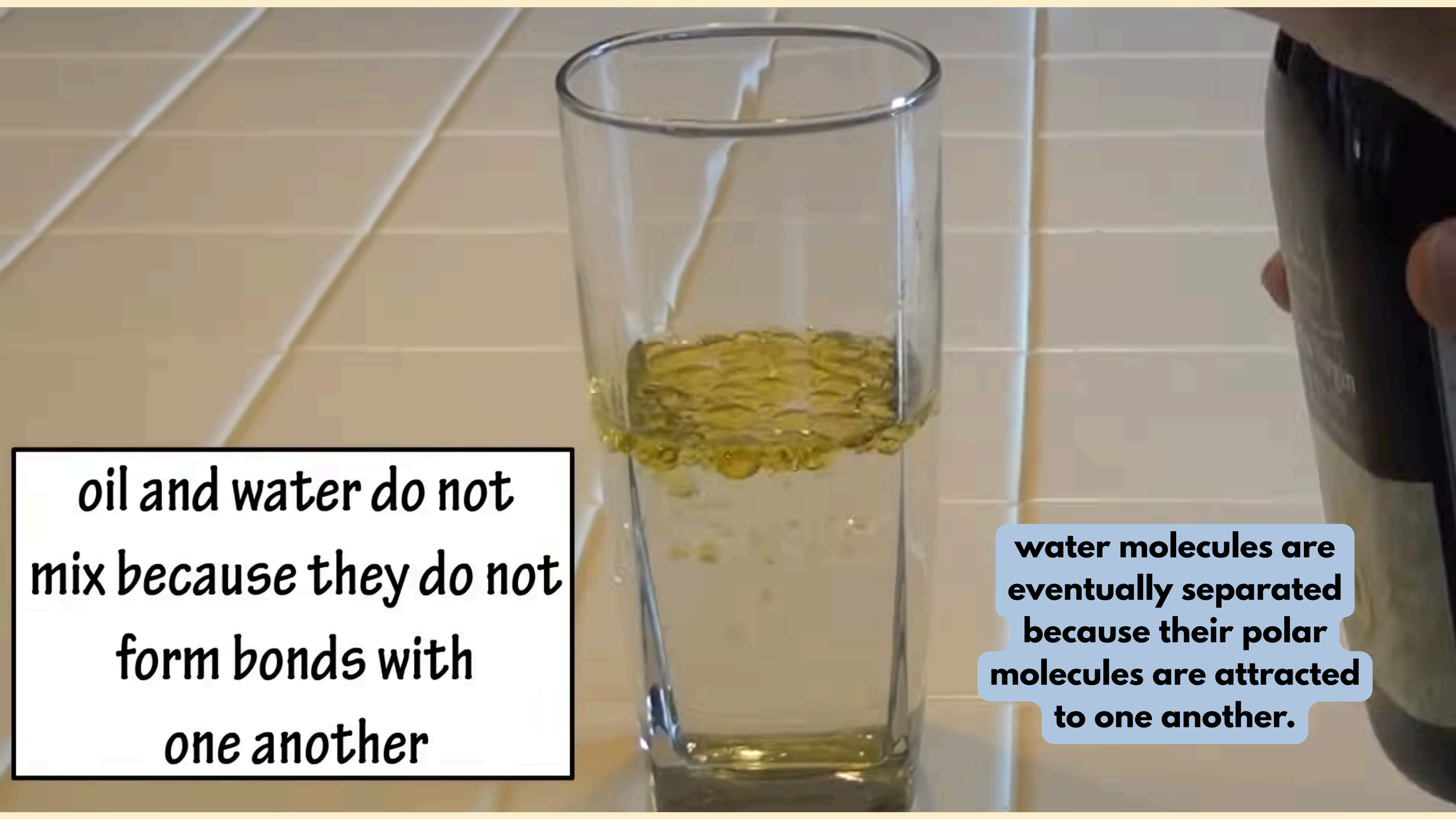
steam



hydrogen bonds



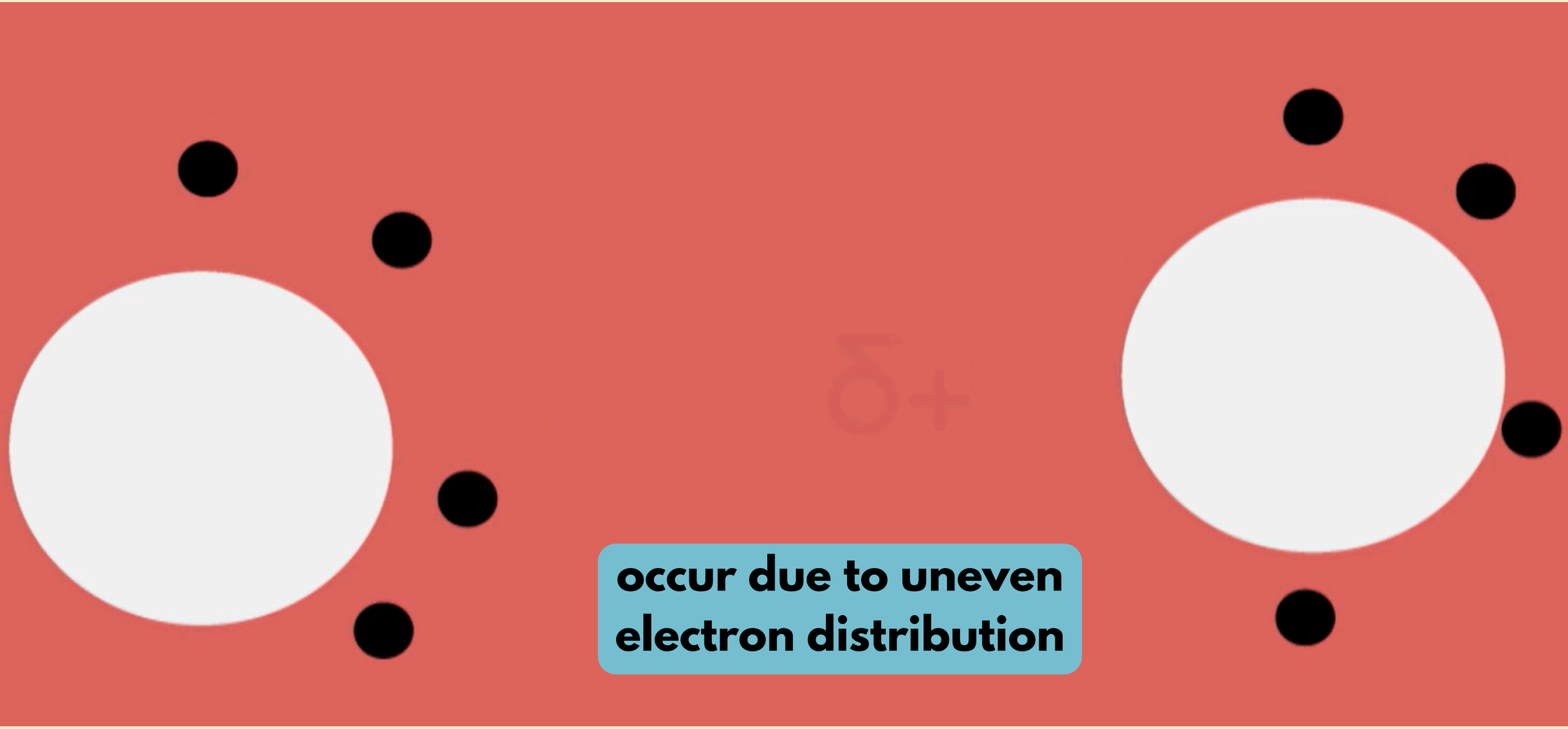
also important in the structure of proteins and nucleic acids

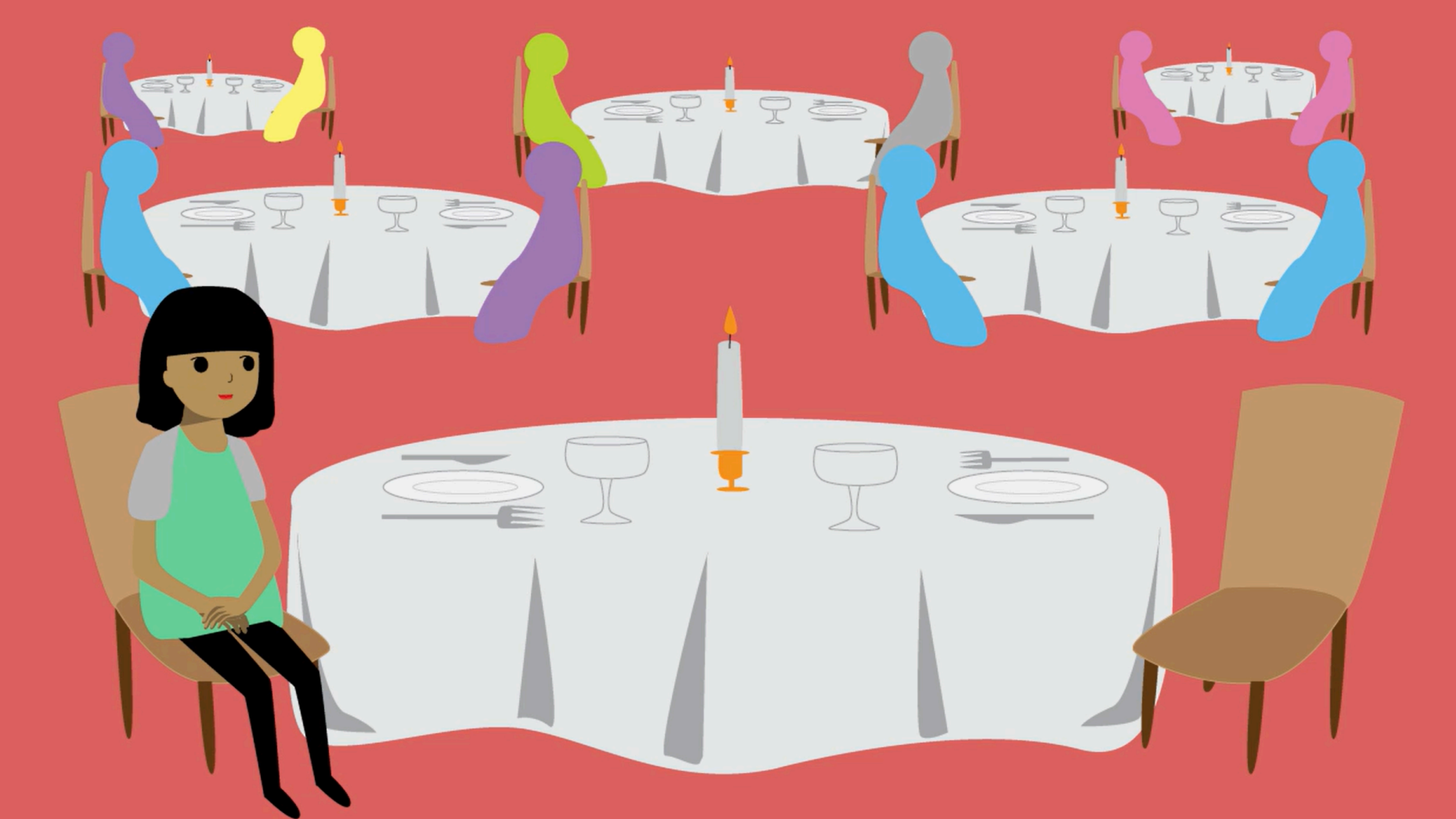

oil and water do not mix because they do not form bonds with one another
water molecules are eventually separated because their polar molecules are attracted to one another.

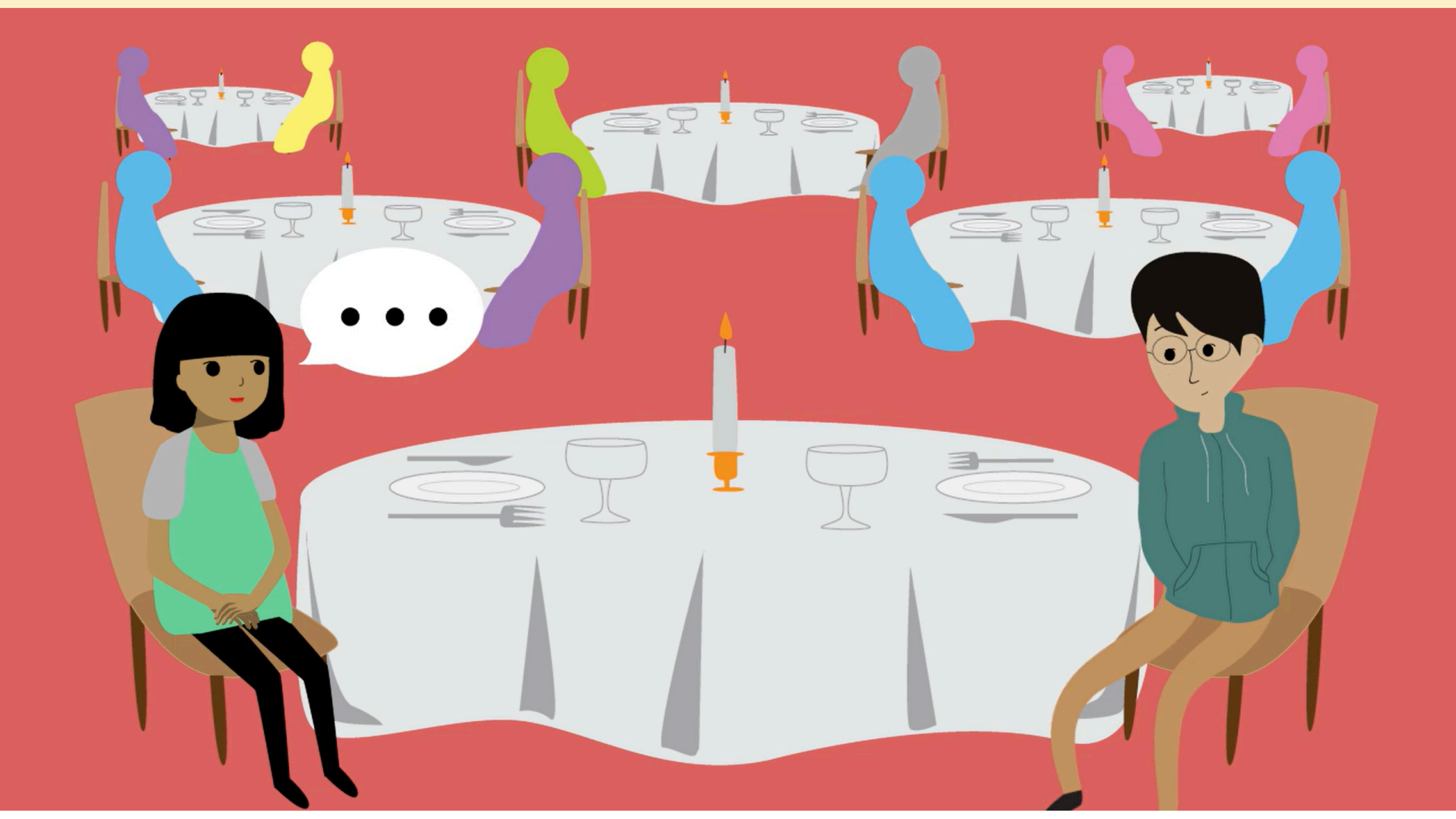
Intermolecular Forces

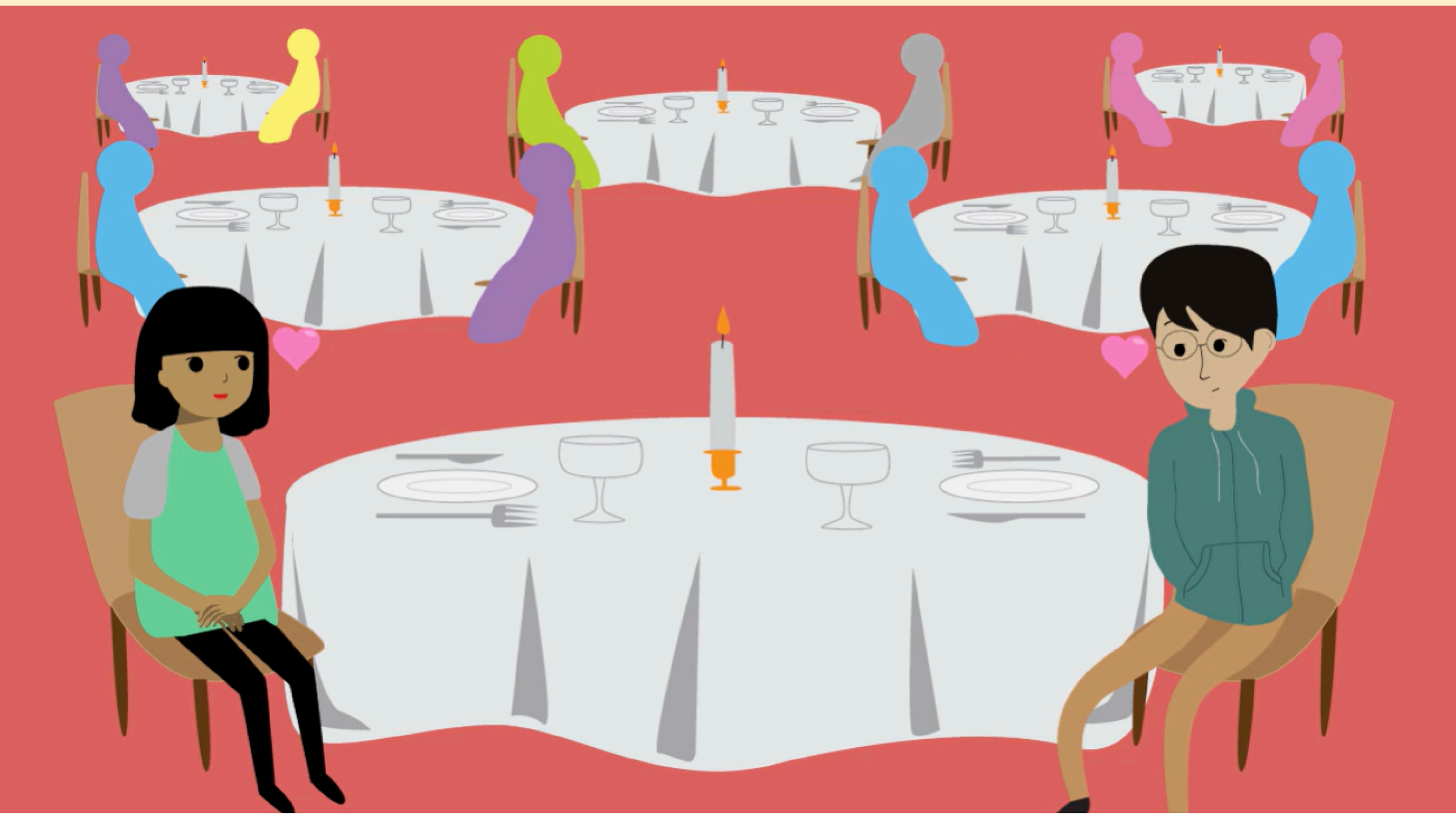
London dispersion force/Induced dipole force

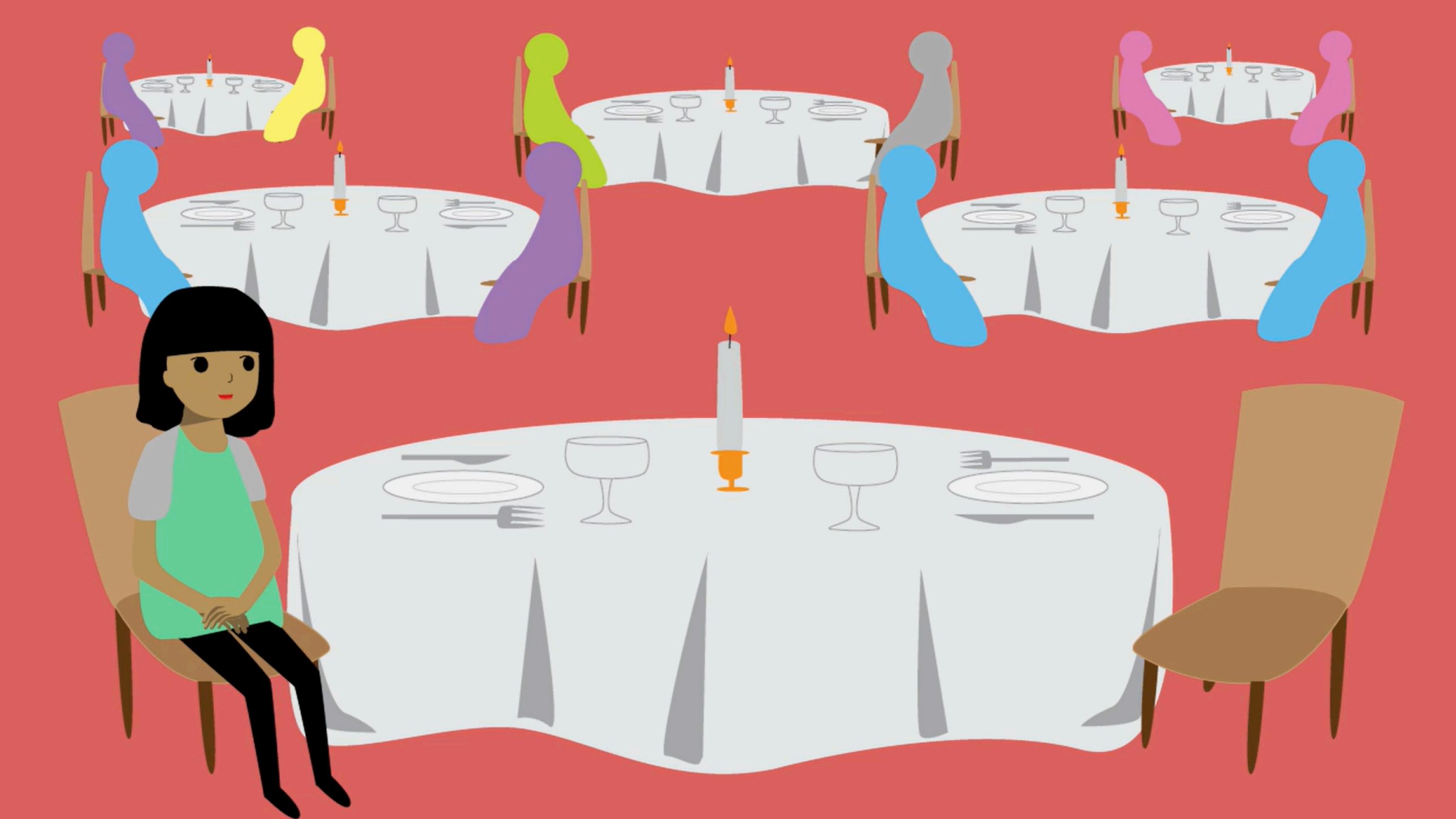
London dispersion force is a weak, temporary attraction between molecules caused by momentary shifts in electron distribution.

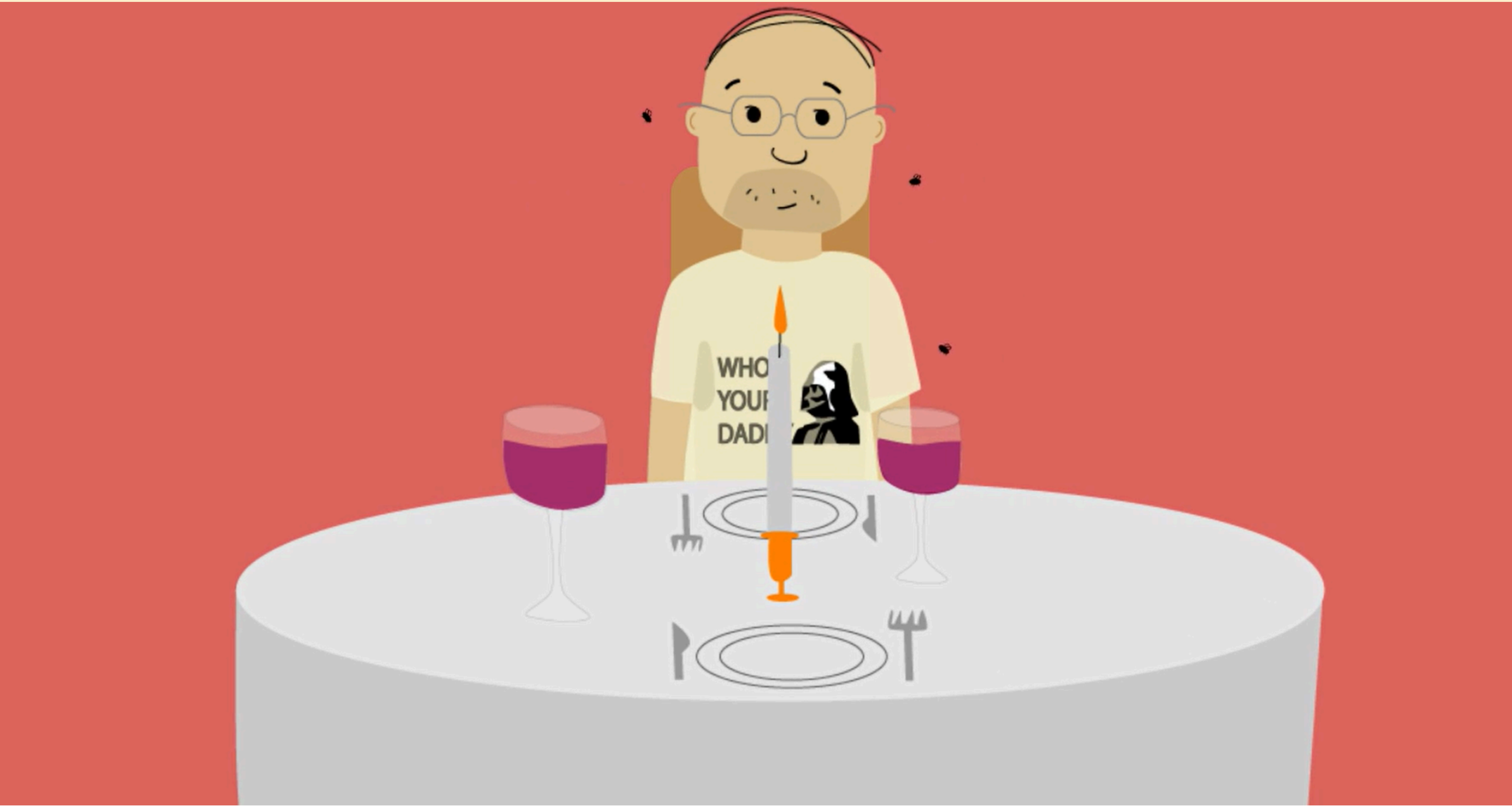

δ+
occur due to uneven
electron distribution

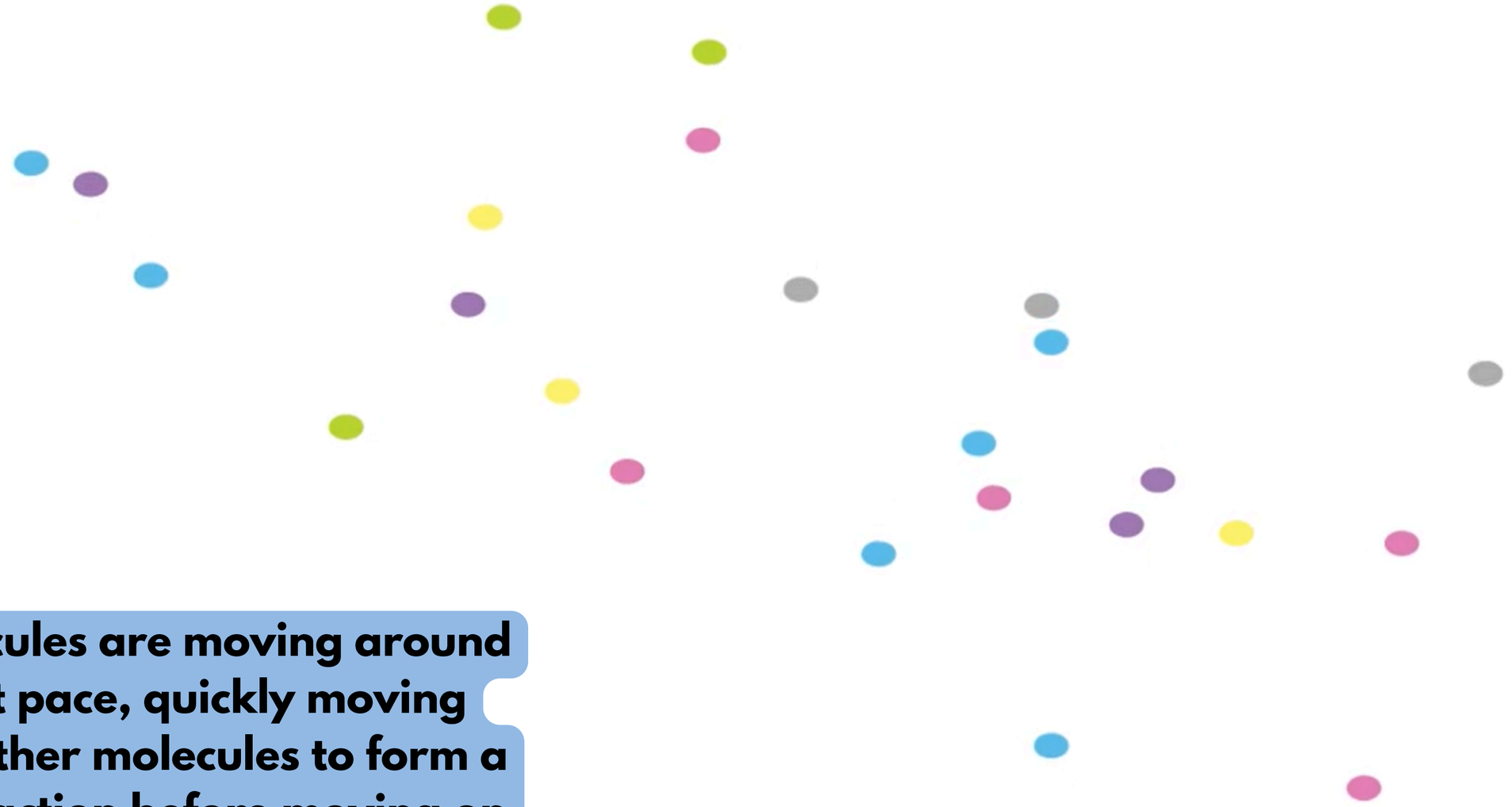








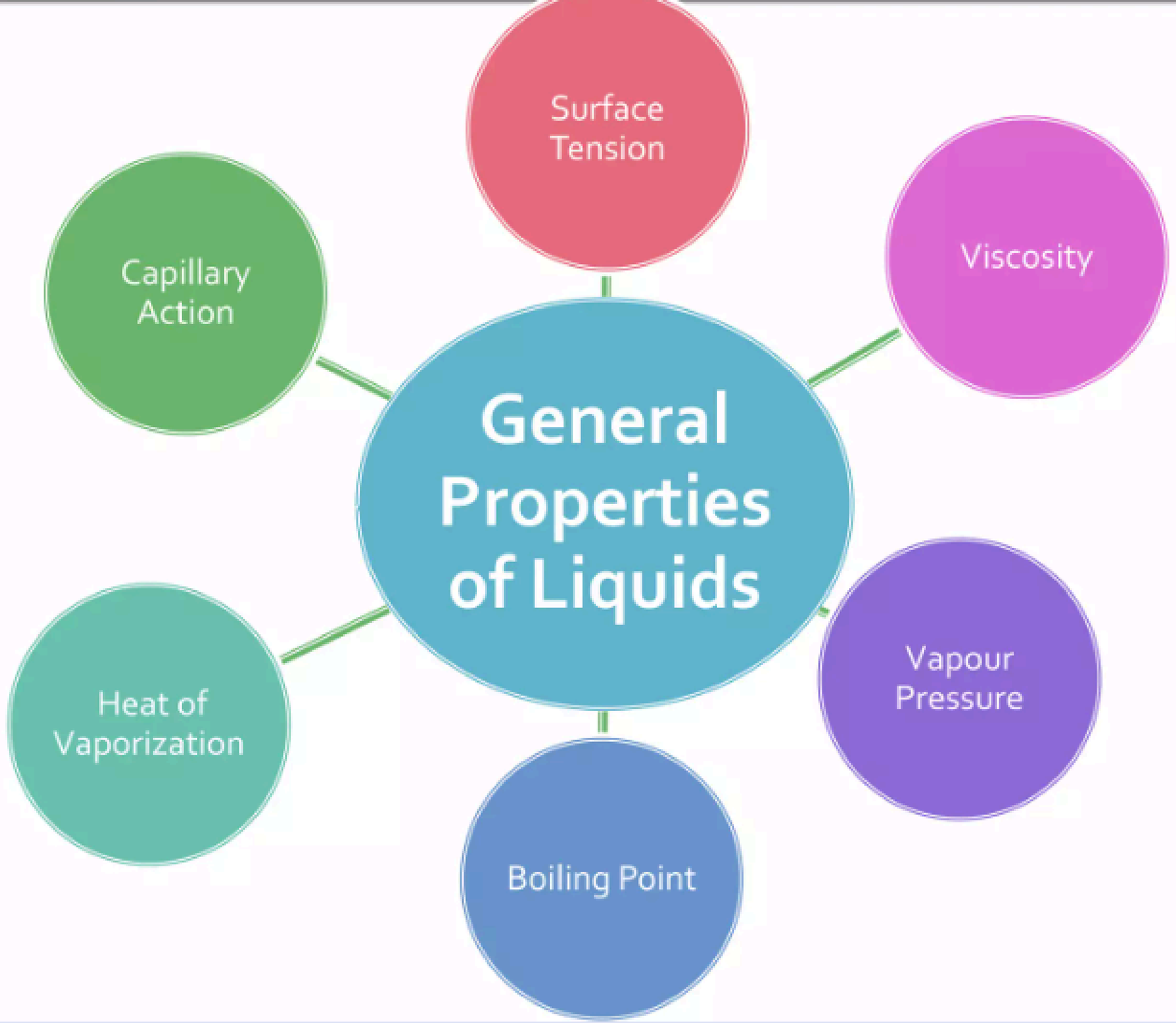

WHO
YOUR
DADD



The molecules are moving around in a fast pace, quickly moving close to other molecules to form a brief attraction before moving on and repeating the cycle again.

LESSON 3

INTERMOLECULAR FORCES AND PROPERTIES OF LIQUID


General Properties of Liquids
Surface Tension
Viscosity
Capillary Action
Vapour Pressure
Heat of Vaporization
Boiling Point

SURFACE TENSION

Surface tension is the liquid's tendency to form a "skin" on the surface due to strong attraction between molecules.

SURFACE TENSION

It happens because of the molecules beside and below them than from the air above.

Why are drops spherical?

CAPILLARY ACTION

Water rises in thin tubes because it sticks to the tube walls (adhesion) and pulls other water molecules upward (cohesion).

CAPILLARY ACTION

EXAMPLES:

- **Water rising in plant stems and roots**
- **Paper towels or tissue absorbing water**
- **Liquid moving up a thin glass capillary in the lab**

VISCOSITY

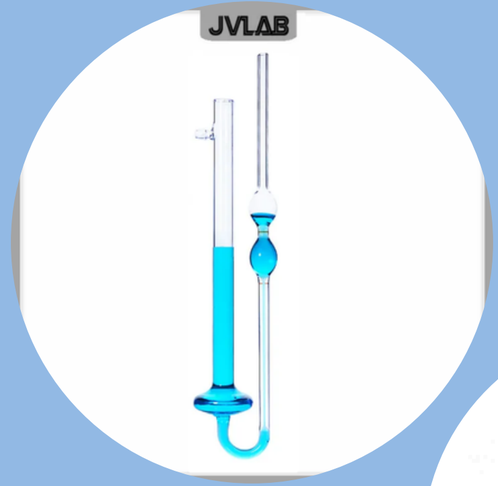
It is a measure of a fluid's resistance to flow.

- **A high-viscosity fluid flows slowly (like honey or syrup).**
- **A low-viscosity fluid flows easily (like water or alcohol).**

VISCOSITY

How Is Viscosity Measured?

- Capillary viscometer (measures flow time through a tube)
- Rotational viscometer (measures torque needed to rotate a spindle in fluid)
- Falling sphere viscometer (based on a ball's falling speed)



VISCOSITY

Why Is Viscosity Important?

Viscosity matters in many fields:

- Engineering: lubrication, pump design, hydraulics
- Medicine: blood viscosity affects circulation
- Food industry: texture of sauces, syrups, dairy
- Earth science: lava flow patterns

VAPOUR PRESSURE

Vapor pressure is the pressure exerted by a vapor when it is in equilibrium with its liquid at a given temperature.

Liquids with weaker intermolecular forces → more molecules escape → higher vapor pressure

HEAT OF VAPORIZATION

Heat of vaporization is the energy needed to overcome the intermolecular forces that hold liquid molecules together so they can escape into the gas phase.

During vaporization:

- **Temperature stays constant**
- **Heat is used to break intermolecular attractions, not to increase temperature**

BOILING POINT

The boiling point of a liquid is the temperature at which its vapor pressure becomes equal to the external (atmospheric) pressure.

STRUCTURE AND PROPERTIES OF WATER

- **Colorless, odorless, and tasteless liquid**
- **It turns to ice, solid form at 0°C and 1 atm.**
- **At 100°C, it becomes a gas, commonly known as steam.**

UNIQUE PROPERTIES OF WATER

1. Water is a good solvent.
2. Water has a high specific heat.

<u>Specific heat</u> is the amount of heat or energy needed to raise the temperature of one gram of a substance by 1° C.

3. The boiling point of water unusually high.

ACTIVITY 2

1. **A tall plant can move water from roots to leaves without a mechanical pump.**

a. Which water properties help this happen? (choose at least two)

b. Explain the process using the terms cohesion, adhesion, and capillary action.